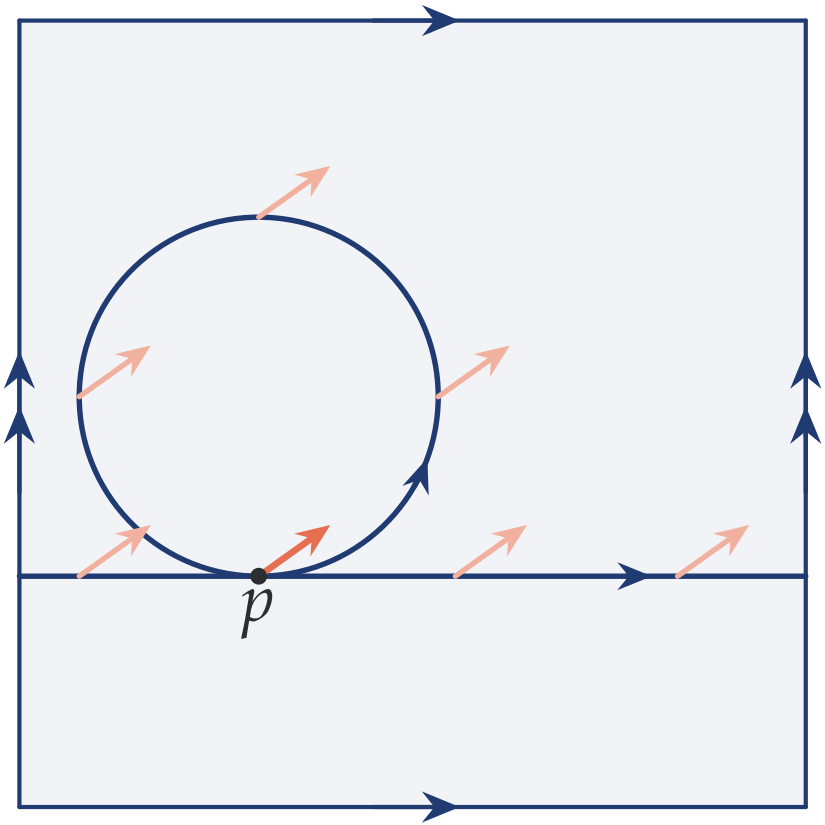
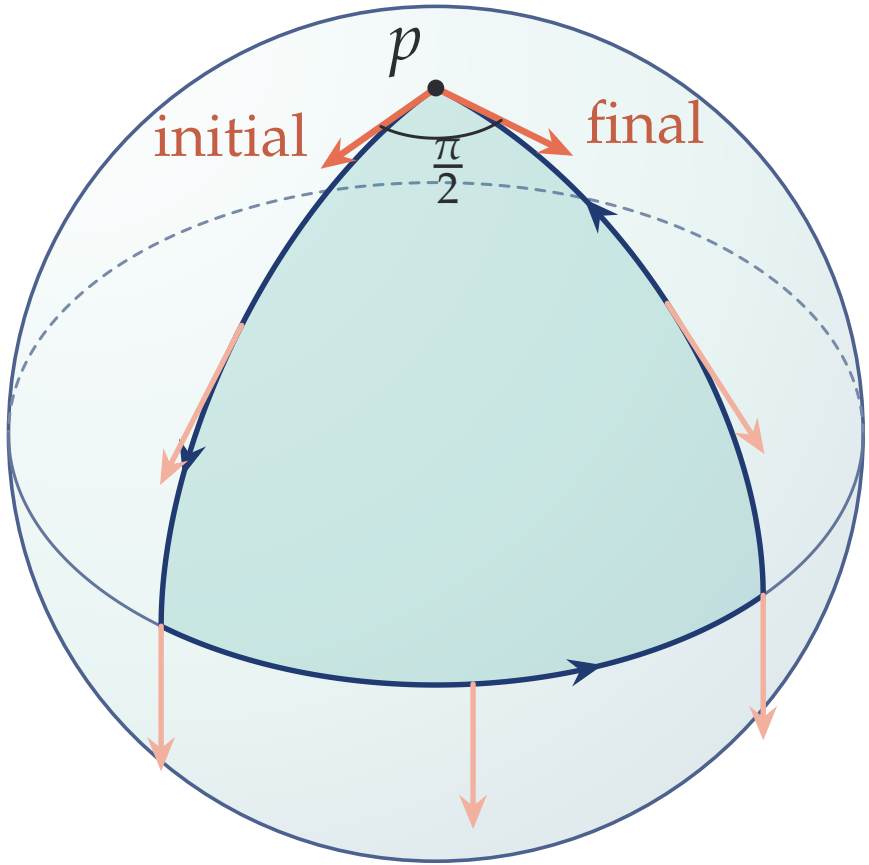
(a) flat torus T^2



$\mathrm{Hol}(T^2) = \{\mathbb{1}\}$

(b) sphere S^2



$\mathrm{Hol}(S^2) = SO(2)$